Financial Risk Analysis with Python- Barclays

Project overview:

You are hired as a Financial Data Analyst by Barclays, a major global banking institution. As digital transactions rise and customers diversify their account usage, the executive team wants to uncover patterns in customer financial behavior, account performance, and transaction risks to improve customer service and minimize financial risk.

The leadership is particularly interested in understanding:

- How do customers interact with different account types (e.g., savings, current, credit)?
- What are the trends in debit vs. credit transactions across segments?
- Can we identify accounts with unusually high risk or inconsistent financial activity?
- What transaction behaviors correlate with lower account balances or overdraft incidents?

Objective: Build a complete Customer Financial Behavior and Risk Analysis Report using Python for data handling. This report should clean and consolidate Barclays' customer account data, uncover transaction insights, and help decision-makers align customer strategies with usage trends and financial risk indicators.

(If you're unfamiliar with concepts, or domain-specific terms, feel free to take the help of Google or AI tools like ChatGPT or Gemini to understand them better and apply them effectively in your project.)

Dataset: Barclays Financial Transactional Data

Dataset Overview: This dataset provides a transactional summary of Barclays' customer accounts, including account numbers, transaction dates, transaction types (credit/debit), transaction amounts, account types, and available balances.

Column Name	Description
TransactionID	Unique identifier for each transaction
CustomerID	Unique identifier for each customer
AccountID	Unique identifier for each bank account
AccountType	Type of bank account (e.g., Savings, Loan, Credit)
TransactionType	Nature of transaction (e.g., Deposit, Withdrawal, Transfer, Payment)
Product	Financial product involved (e.g., Credit Card, Mutual Fund, Home Loan)
Firm	Name of the Firm where the transaction occurred
Region	Geographical region of the branch (e.g., East, West, North)
Manager	Branch manager responsible for the transaction or customer
TransactionDate	Date when the transaction occurred
TransactionAmount	Amount involved in the transaction
AccountBalance	Account balance after the transaction
RiskScore	Risk score associated with the customer or transaction
CreditRating	Credit rating score of the customer
TenureMonths	Duration (in months) the customer has held the account or relationship

Tasks to be performed:

(Please refrain from using AI to perform the tasks mentioned below, as it will only provide generic solutions.)

Task 1: Data Cleaning and Formatting (10 Marks)

- Remove/treat any special characters or non-numeric entries from financial fields.
- Convert currency amounts into numerical format.
- Validate and format date columns.
- Ensure account types and transaction categories are standardized.

Task 2: Descriptive Transactional Analysis (20 Marks)

- Calculate monthly and yearly summaries of total credits, debits, and net transaction volume.
- Plot trends in total credits vs. debits over time.
- Identify top and bottom performing accounts based on net inflow.
- Identify and flag accounts as dormant or inactive if there is a gap of two months or more between consecutive transactions.

Task 3: Customer Profile Building (20 Marks)

- Group accounts by activity levels: High, Medium, Low based on transaction frequency on your analysis and rubrics. Do not forget to mention the rubric in the headings.
- Segment customers by average balance and transaction volume.
- Create profiles for:

- High-net inflow accounts
- High-frequency low-balance accounts
- Accounts with negative or near-zero balances

Task 4: Financial Risk Identification (10 Marks)

- Track accounts with frequent large withdrawals or overdrafts.
- Calculate balance volatility using standard deviation or coefficient of variation.
- Use IQR or z-score methods to detect anomalies.
- Highlight customers with irregular or suspicious transaction behavior.

Task 5: Visualisation (10 Marks)

- Conduct extensive exploratory data analysis with attractive visualizations for your findings

Task 6: Hypothesis Testing (10 Marks)

- Test whether high-volume transaction accounts have statistically higher average balances than low-volume accounts.
- Conduct hypothesis testing based on segmentation.

Task 7: Video Presentation (20 Marks)

(The summary should be in your own words and must not be generated using AI. Please don't write a script and read it aloud. Use charts from your analysis as visual aids in the video.)

- Record a short video of 5 mins summarizing findings and insights.
- Highlight what drives customer transaction behavior and financial risk.
- Discuss data-backed recommendations for customer engagement or monitoring.

Note:

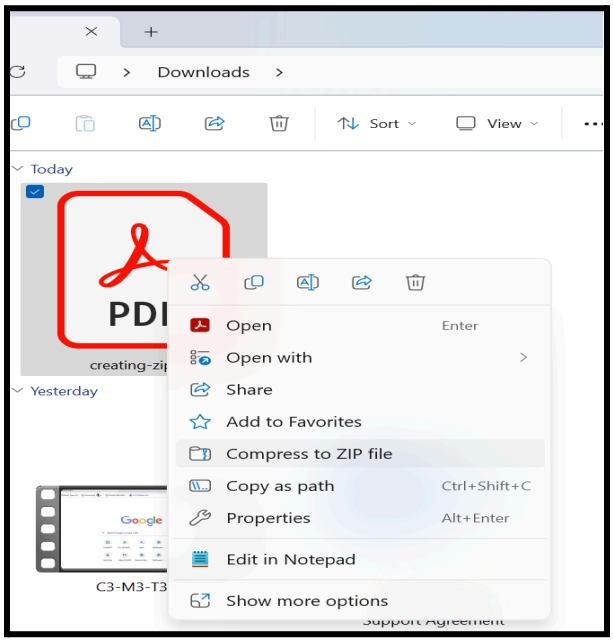
- Place all code files, and summary PDF in one folder and then zip that folder. Follow basic coding ethics (headlines, pointers for insights and comments wherever required)
- Plagiarism will result in a penalty, including possible project disqualification.
- The project will be evaluated based on the quality of analysis and visualizations, depth of insights, feasibility of recommendations, clarity of explanations, and adherence to instructions and deliverables.
- If you set your own criteria (e.g., for flagging risk years or segmenting periods), clearly mention them. Marks will be given according to the specified criteria if they are acceptable.
- Remember to keep the video length less than 5 minutes with your face clearly visible. Add the video link in the code file or in the document.

Submission Guidelines:

- Save the code file, and summary in a PDF and then convert it into a zipped (.zip) folder. **(Please note, the drivelink for the video created should also be added in the PDF itself.)**
- Upload the zipped folder on your respective dashboard.
- Failure to comply with submission guidelines will result in no grading/0 marks.

How to ZIP a PDF file:

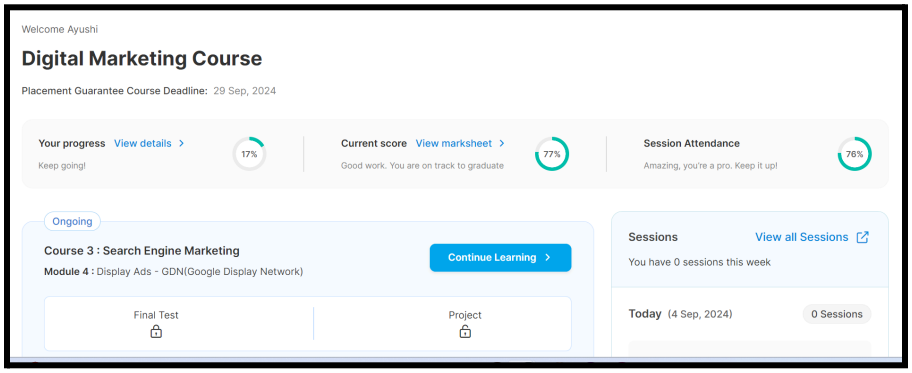
- Put all of the documents you want to compress (or just one) into a new folder.
- Right click on that folder.



- Select the “Compress to ZIP file” option and then click “Compressed (Zipped) folder.”
- A new .ZIP file will be created that contains your document(s).

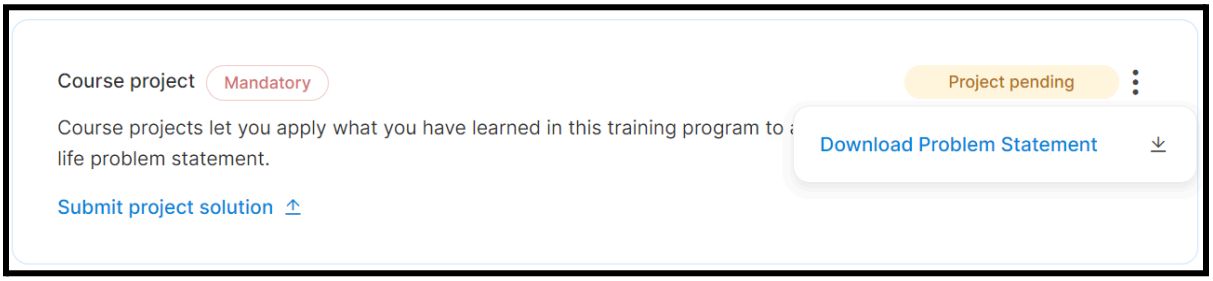
In order to submit the projects please follow the following steps:

1. Click on “ Your progress View details” after logging into your dashboard.



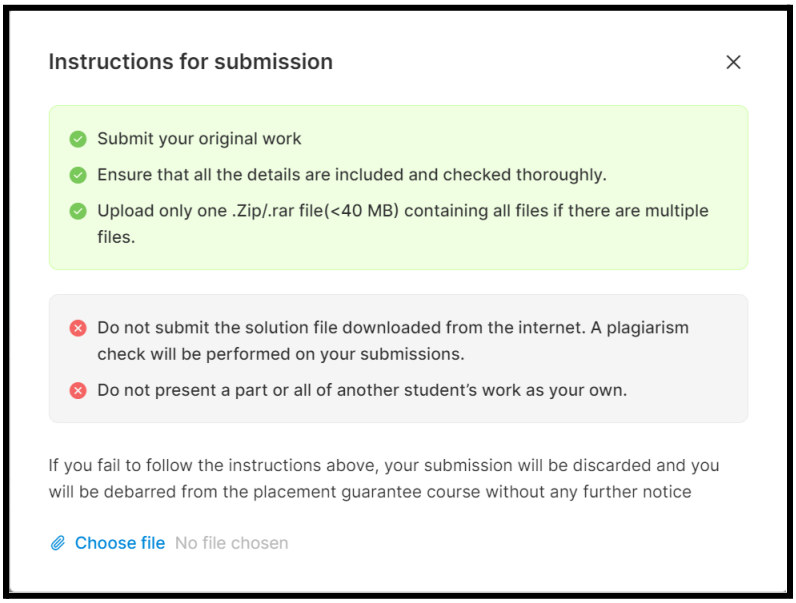
2. Next, click on the tab for the specific child course for which you want to download the problem statement. Then, scroll down to find the "**Course Project**" section.

3. Now, click on the three dots on the right-hand side of the "Course Project" tab to select **"Download Problem Statement."**



4. Please follow the guidelines (screenshot is shared below) provided in the project to ensure correct submissions. Then, click on "**Upload Project Solution**" to submit

your work.


Instructions for submission
Submit your original work
Ensure that all the details are included and checked thoroughly.
Upload only one .Zip/.rar file(<40 MB) containing all files if there are multiple files.
Do not submit the solution file downloaded from the internet. A plagiarism check will be performed on your submissions.
Do not present a part or all of another student's work as your own.
If you fail to follow the instructions above, your submission will be discarded and you will be debarred from the placement guarantee course without any further notice
Choose file
No file chosen